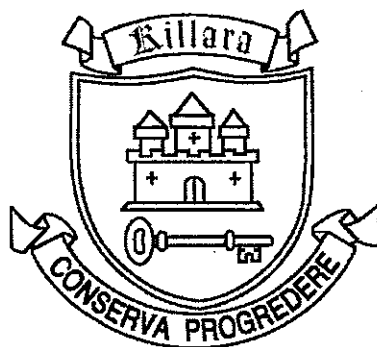


Te 10ciii



Killara High School

2010

TERM 1 EXAMINATION

Monday, 3rd March

Mathematics

General Instructions**Total marks – 120**

- Reading time – 5 minutes
- Working time – 3 hours
- Start each question in a new booklet
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Attempt Questions 1–10
- All questions are of equal value

Question 1 (12 Marks)

Start a new booklet

Marks

- (a) Evaluate correct to three significant figures,

2

$$\sqrt{\frac{3^2 + 12^2}{231 - 12^2}}$$

- (b) Factorise: $2x^3 + 16$

2

- (c) Solve: $\frac{a-5}{3} - \frac{a+1}{4} = 5$

2

- (d) Find integers a and b such that $\frac{2}{3-\sqrt{5}} = a + b\sqrt{5}$

2

- (e) Differentiate $x^5 + 3x^{-1}$

2

- (f) Find the values of x for which $|x+1| \leq 5$

2

Question 2 (12 Marks)

Start a new booklet

Marks

- (a) Sketch the graph of $y = 2x + 3$ on the cartesian plane, showing clearly the x and y intercepts. 2

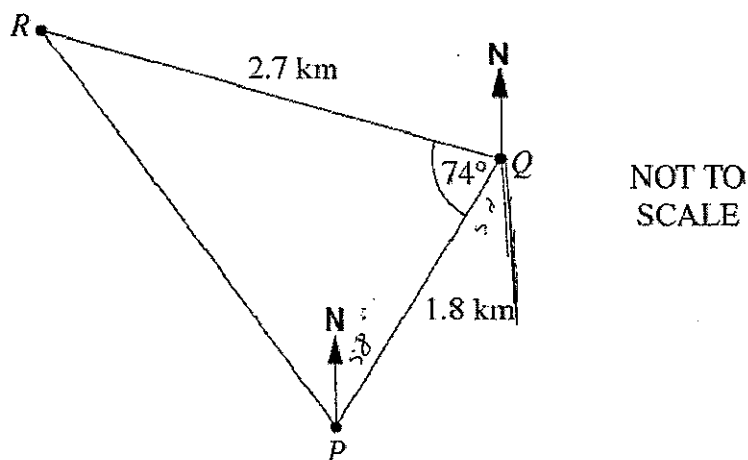
- (b) Differentiate:

(i) $y = \frac{3x}{5-x}$ 2

(ii) $y = x^2 \sqrt{3x-7}$ 2

(iii) $y = \frac{5}{(3x-2)^8}$ 2

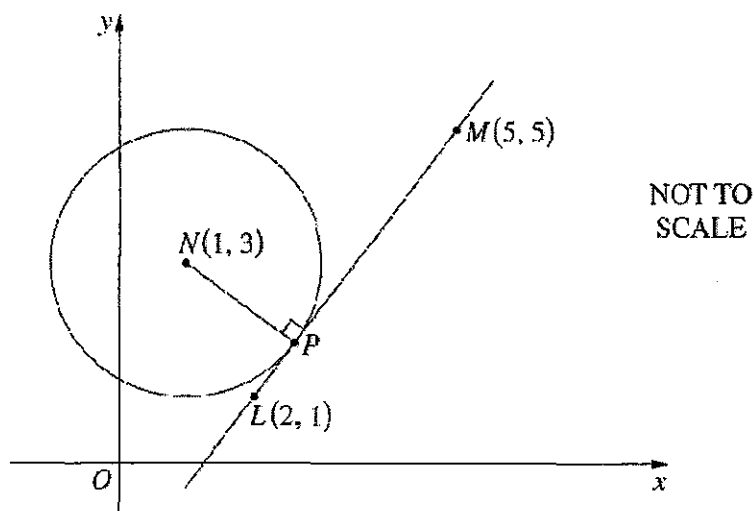
- (c) A yacht race follows the triangular course as shown in the diagram. The course from P to Q is 1.8 km on a bearing of 058° . At Q the course changes direction. The course from Q to R is 2.7 km and $\angle PQR = 74^\circ$



- (i) What is the bearing of R from Q ? 1
- (ii) What is the distance from R to P ? 2
- (iii) The area inside this course is set as a 'no-go' zone for other boats while the race is on. 1

What is the area of this 'no go' zone?

- | Question 3 | (12 Marks) | Start a new booklet | Marks |
|------------|--|---------------------|-------|
| (a) | Find the coordinates of the midpoint of the interval joining the points (2, 1) and (4, -3). | | 1 |
| (b) | Find the equation of the tangent to the curve $y = x^2 - 2x$ at the point (-1, 3). | | 2 |
| (c) | Find the exact value of θ such that $\sqrt{3} \cos \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ | | 2 |
| (d) | Shade the region in the number plane defined by $y \geq 0$ and $y \leq 9 - x^2$ | | 2 |
| (e) | | | |



The circle in the diagram has centre N. The line LM is a tangent to the circle at P.

- | | | |
|-------|---|---|
| (i) | Find the equation of LM and leave your answer in the form $ax + by + c = 0$ | 2 |
| (ii) | Find the distance NP. | 2 |
| (iii) | Find the equation of the circle | 1 |

Question 4. (12 Marks)

Start a new booklet

Marks

- (a) Differentiate, from first principles, $f(x) = 2 - x^2$

2

- (b) The equation of a parabola is $x^2 = 8(y + 3)$

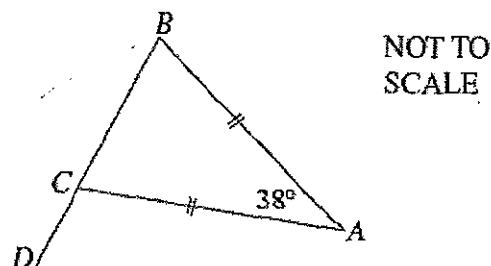
- (i) Find the coordinates of the vertex of the parabola

1

- (ii) Find the equation of the directrix of the parabola

2

- (c)



In the diagram, ABC is an isosceles triangle with $AB = AC$ and $\angle BAC = 38^\circ$.

Copy or trace the diagram into your answer booklet.

Find the size of $\angle ACD$. You must give reasons for your answer.

2

- (d) Evaluate : $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

2

- (e) Find the values of k for which the quadratic equation

3

$$x^2 - (k + 4)x + (k + 7) = 0$$

has equal roots

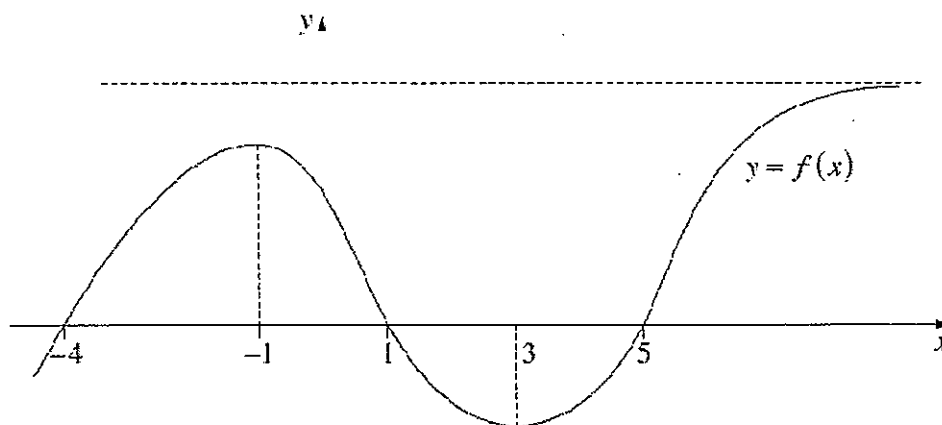
Question 5. (12 Marks)	Start a new booklet	Marks
(a) Solve the equation $m^2 - m - 1 = 0$, correct to three decimal places		2
(b) Find the exact value of $\sec 210^\circ$		2
(c) Find the equation of the normal to the curve $x^2 = 8y$ at the point where $x = 4$.		4
(d) Let α and β be the roots of the equation $x^2 - 5x + 2 = 0$. Find the values of:		
(i) $\alpha + \beta$		1
(ii) $\alpha \beta$		1
(iii) $(\alpha + 1)(\beta + 1)$		2

Question 6. (12 Marks)	Start a new booklet	Marks
(a) Show that $1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x}$ for $x \neq -1$.		2
(b) The roots of the quadratic equation $px^2 - x + q = 0$ are -1 and 3. Find p and q .		2
(c) Solve: $4^x - 5(2^{x+1}) + 16 = 0$		3
(d) (i) Given $f(x) = \frac{x}{x-1}$, evaluate $f\left(\frac{1}{2}\right) - f(-2)$		1
(ii) Show that $f(x) = \frac{x}{x-1}$ is neither an odd function nor an even function.		2
(e) Prove $(1 - \cos \theta)(1 + \sec \theta) = \sin \theta \tan \theta$		2

Question 7 (12 Marks)	Start a new booklet	Marks
(a) For what values of x is $2x^2 \geq x$?		2
(b) Find a primitive function of $6 - x^{-3}$		2
(c) (i) Find the sum of the interior angles of a regular 11 sided polygon.		1
(ii) How large is each exterior angle to the nearest minute?		1
(d) Express $5x^2 + 2x - 3$ in the form $A(x+1)^2 + B(x+1) + C$		2
(e) For the parabola defined by $x^2 + 4x - 8y + 12 = 0$ find the coordinates of the focus.		2
(f) For all x in the domain $-2 \leq x \leq 2$, a function, $f(x)$ is such that $f(x) < 0$, $f'(x) > 0$ and $f''(x) < 0$. Sketch a possible $y = f(x)$ in this domain.		2

Question 8. (12 Marks)

Start a new booklet

Marks**(a)**

The diagram shows the graph of a function $y = f(x)$

- | | | |
|-------|--|---|
| (i) | Sketch the graph $y = f'(x)$. | 2 |
| (ii) | For which values of x is the derivative, $f'(x)$, negative? | 1 |
| (iii) | What happens to $f'(x)$ for large values of x ? | 1 |

(b) Consider the curve given by $y = 7 + 4x^3 - 3x^4$

- | | | |
|-------|--|---|
| (i) | Find the coordinates of the two stationary points. | 2 |
| (ii) | Determine the nature of the stationary points | 2 |
| (iii) | Find the point of inflexion. | 1 |
| (iv) | Sketch the curve for the domain $-1 \leq x \leq 2$ | 2 |
| (v) | State the greatest value of y in the domain. | 1 |

Question 9. (12 Marks)

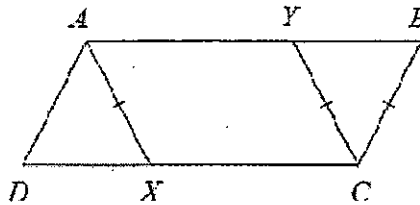
Start a new booklet

Marks

- (a) If $\tan \theta = -\frac{4}{9}$ and $\cos \theta$ is positive, find the exact value of $\sin \theta$

2

(b)



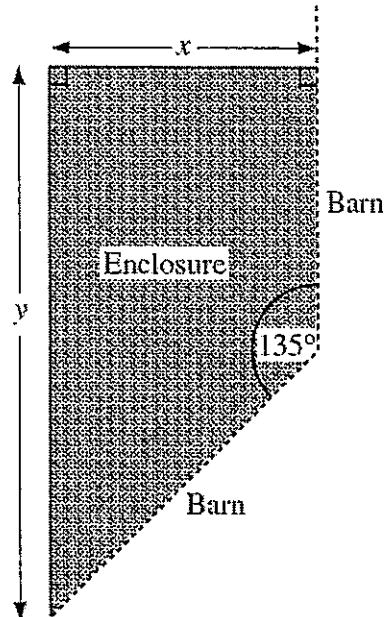
$ABCD$ is a parallelogram. The point X lies on CD , the point Y lies on AB , and $AX = CY = BC$, as shown in the diagram.

Copy the diagram into your writing booklet.

- (i) Explain why $\angle ADX = \angle CBY$ **1**
- (ii) Show that $AD = AX$ **1**
- (iii) Show that triangles ADX and CBY are congruent. **2**
- (iv) Hence prove that $AYCX$ is a parallelogram. **1**

QUESTION 9 IS CONTINUED ON THE NEXT PAGE

- (c) An enclosure is to be built adjoining a barn, as in the diagram. The walls of the barn meet at 135° , and 117m of fencing is available for the enclosure, so that $x + y = 117$ where x and y are as shown in the diagram.



- (i) Show that the shaded area of the enclosure in square metres is given by 2

$$A = 117x - \frac{3}{2}x^2$$

- (ii) Show that the largest area of the enclosure occurs when $y = 2x$. 3

Question 10. (12 Marks)

Start a new booklet

Marks

(a) Simplify: $\frac{1-x^2}{x-1}$

1

(b) Plot the points $A(0, 1)$, $B(4, 0)$ and $C(0, 2)$ on a number plane in your answer booklet.

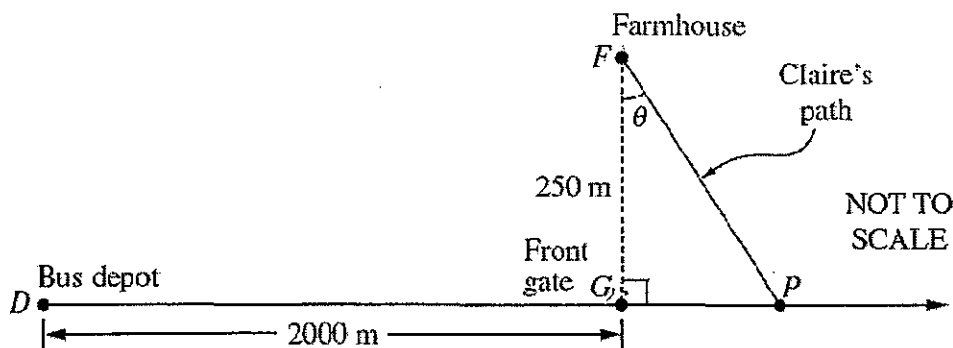
(i) Show that the length CB is twice that of CA

2

(ii) A point $P(X, Y)$ moves such that the length of PB is twice that of PA . Show that the locus of P is a circle, and determine its centre and radius.

3

(c)



The diagram shows a farmhouse F that is located 250 m from a straight section of road. The road begins at the bus depot D , which is situated 2000 m from the front gate G of the farmhouse. The school bus leaves the depot at 8 am and travels along the road at a speed of 15 m s^{-1} . Claire lives in the farmhouse, and she can run across the open paddock between the house and the road at a speed of 4 m s^{-1} . The bus will stop for Claire anywhere on the road, but will not wait for her.

Assume that Claire catches the bus at the point P on the road where $\angle GFP = \theta$.

(i) Find two expressions in terms of θ , one expression for the time taken for the bus to travel from D to P and the other expression for the time taken by Claire to run from F to P .

2

(ii) What is the latest time that Claire can leave home in order to catch the bus?

4

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

KHS.

1/2 July. Mathematics 2010.

Ques 1.

SOLUTIONS.

$$\begin{aligned} a) \sqrt{\frac{3^2 + 12^2}{2 \cdot 31 + 12^2}} &= \sqrt{\frac{9 + 144}{231 + 144}} \\ &= \sqrt{\frac{153}{375}} \\ &= \sqrt{1075862069} \\ &= 1.35 \text{ s.f.} \end{aligned}$$

$$\begin{aligned} f) |x+1| &\leq 5 \\ -5 &\leq x+1 \leq 5 \\ -6 &\leq x \leq 4. \end{aligned}$$

$$\begin{aligned} b) 2x^3 + 16 &= 2(x^3 + 8) \\ &= 2(x+2)(x^2 + 2x + 4) \end{aligned}$$

$$c) \frac{a-5}{3} - \frac{a+1}{4} = 5$$

$$4(a-5) - 3(a+1) = 60$$

$$4a - 20 - 3a - 3 = 60$$

$$a - 23 = 60$$

$$a = 83$$

$$d) \frac{2}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}} = \frac{6+2\sqrt{5}}{9-5}$$

$$= \frac{6+2\sqrt{5}}{4}$$

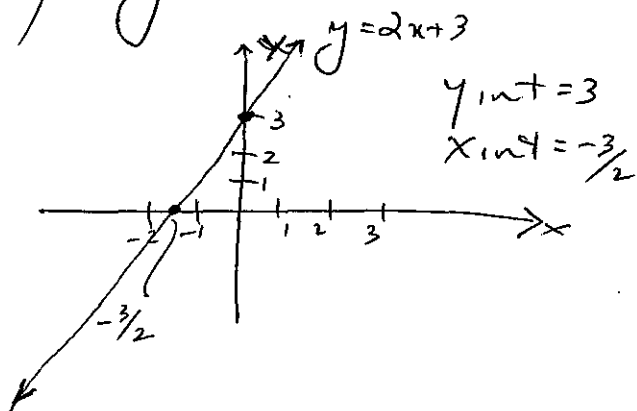
$$= \frac{3}{2} + \frac{\sqrt{5}}{2}$$

$$\therefore a = \frac{3}{2}, b = \frac{1}{2}$$

$$e) \frac{d}{dx} x^5 + 3x^{-1} = 5x^4 - 3x^{-2}$$

Ques 2

a) $y = 2x + 3$



c) iii) $A = \frac{1}{2} \times 2.7 \times 1.8 \times \sin 74^\circ$

$= 2.033586592$
 $\approx 2.034 \text{ km}^2$

b) i) $\frac{d}{dx} \frac{3x}{5-x} = \frac{(5-x) \cdot 3 - (3x) \cdot (-1)}{(5-x)^2}$

$= \frac{15 - 3x + 3x}{(5-x)^2}$

$= \frac{15}{(5-x)^2}$

ii) $y = x^2 \sqrt{3x-7}$

$\frac{d}{dx} = \sqrt{3x-7} \cdot 2x + x^2 \cdot \frac{1}{2} (3x-7)^{-\frac{1}{2}} \cdot 3$

$= 2x \sqrt{3x-7} + \frac{3x^2}{2\sqrt{3x-7}}$

iii) $y = \frac{5}{(3x-2)^8} = 5(3x-2)^{-8}$

$y' = -8 \cdot 5 (3x-2)^{-9} \cdot 3$

$= \frac{-120}{(3x-2)^9}$

c) i) R from $Q = 180 + 58 + 74$
 $= 312^\circ \text{ T}$

ii) $RP^2 = 1.8^2 + 2.7^2 - 2 \times 1.8 \times 2.7 \times \cos 74^\circ$
 $= 7.850804902$
 $\approx 2.8 \text{ km}$

Ques 3

a) $(2, 1)$ and $(4, -3)$

$$m.p = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{2+4}{2}, \frac{1+(-3)}{2} \right)$$

$$= \left(\frac{6}{2}, \frac{-2}{2} \right)$$

$$= (3, -1)$$

b) $y = x^2 - 2x$ $(-1, 3)$

$$y' = 2x - 2$$

$$m = 2x - 1 - 2$$

$$= -4$$

$\therefore$ tangent

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -4(x - -1)$$

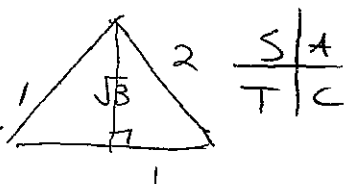
$$y - 3 = -4x - 4$$

$$4x + y + 1 = 0$$

c) $\sqrt{3} \cos \theta = 1$ $0 \leq \theta \leq 360^\circ$

$$\cos \theta = \frac{1}{\sqrt{3}}$$

basic $\theta = 54^\circ 44'$

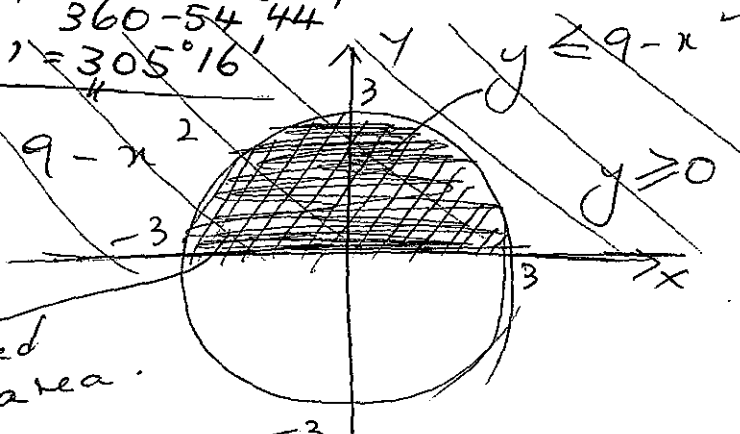


* $\therefore \theta = 54^\circ 44'$ $360 - 54^\circ 44' = 305^\circ 16'$

d) $y \geq 0$

$y \leq 9 - x^2$

required area.



e) i) $L(2, 1)$ $M(5, 5)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 1}{5 - 2}$$

$$= \frac{4}{3}$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{4}{3}(x - 2)$$

$$3y - 3 = 4x - 8$$

$$0 = 4x - 3y - 5$$

ii) perp dist.

$$= \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$= \frac{|4x - 3y - 5|}{\sqrt{4^2 + (-3)^2}}$$

$$= \frac{|4 - 9 - 5|}{\sqrt{25}}$$

$$= \frac{8}{5} \text{ units.}$$

iii) $(x - 1)^2 + (y - 3)^2 = \left(\frac{8}{5}\right)^2$

$$(x - 1)^2 + (y - 3)^2 = \frac{64}{25}$$

Ques 4

a) $y = 2 - x^2$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 - (x+h)^2 - (2 - x^2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 - (x^2 + 2xh + h^2) - 2 + x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-2xh - h^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(-2x - h)}{h}$$

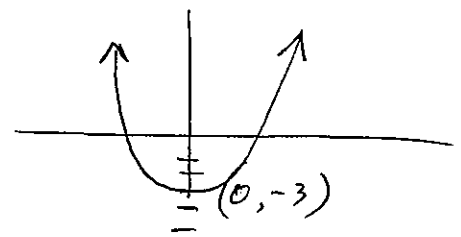
$$= -2x$$

b) $x^2 = 8(y+3)$

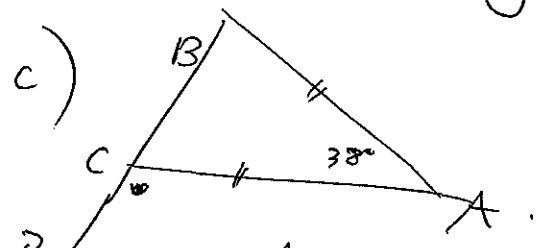
like $(x-h)^2 = 4a(y-k)$

i) $\therefore V = (0, -3)$

ii) $4a = 8 \therefore a = 2$



$\therefore$ directrix is $y = -5$.



$$\angle ACB = \angle ABC = \frac{180 - 38}{2} \text{ (base } \angle \text{ s of } \triangle \text{)}$$

$$= 71^\circ$$

$$\therefore \angle A D C = 180 - 71^\circ \text{ (st. } \angle = 180) = 109^\circ$$

d) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

$$\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2}$$

$$= 4$$

e) $x^2 - (k+4)x + (k+7) = 0$
equal roots

$$\Delta = 0, b^2 - 4ac = 0$$

$$-(k+4)^2 - 4 \cdot 1 \cdot (k+7) = 0$$

$$k^2 + 8k + 16 - 4k - 28 = 0$$

$$k^2 + 4k - 12 = 0$$

$$(k+6)(k-2) = 0$$

$$\therefore k = -6 \text{ or } 2$$

Ques 5

$$a) m^2 - m - 1 = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot -1}}{2 \cdot 1}$$

$$= \frac{1 \pm \sqrt{1+4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

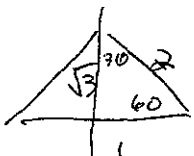
$$= 1.079, -0.412 \text{ (3 d.p.)}$$

$$b) \sec 210^\circ = \frac{1}{\cos 210^\circ}$$

$$\frac{S}{T} \frac{A}{C} = \frac{1}{-\cos 30^\circ}$$

$$= \frac{1}{-\frac{\sqrt{3}}{2}}$$

$$= -\frac{2}{\sqrt{3}}$$



$$c) x^2 = 8y \quad x = 4$$

$$y = \frac{x^2}{8}$$

$$y' = \frac{2x}{8} = \frac{x}{4}$$

$$\text{at } x=4, y' = \frac{4}{4} = 1, \therefore \text{normal has } m = -1$$

$$x=4, y=2$$

equation of normal,

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -1(x - 4)$$

$$y - 2 = -x + 4$$

$$d) x^2 - 5x + 2 = 0$$

$$i) \alpha + \beta = -\frac{b}{a}$$

$$= -\frac{-5}{1}$$

$$= 5$$

$$ii) \alpha\beta = \frac{c}{a}$$

$$= \frac{2}{1}$$

$$= 2$$

$$iii) (\alpha+1)(\beta+1)$$

$$= \alpha\beta + \alpha + \beta + 1$$

$$= 2 + 5 + 1$$

$$= 8$$

$$e) (1 - \cos \theta)(1 + \sec \theta) = \sin \theta \tan \theta$$

$$LHS = 1 + \sec \theta - \cos \theta - \cos \theta \sec \theta$$

$$= 1 + \sec \theta - \cos \theta - \cos \theta \left(\frac{1}{\cos \theta} \right)$$

$$= 1 + \sec \theta - \cos \theta - 1$$

$$= \frac{1}{\cos \theta} - \cos \theta$$

$$= \frac{1 - \cos^2 \theta}{\cos \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta}$$

$$= \frac{\sin \theta \cdot \sin \theta}{\cos \theta}$$

$$= \sin \theta \cdot \tan \theta$$

$$= RHS.$$

Ques 6(e)

Ques 6

$$a) \frac{1 - x + x^2 - 1}{1+x} = \frac{x^3}{1+x}$$

$$LHS = \frac{(1-x+x^2)(1+x) - 1}{1+x}$$

$$= \frac{\cancel{1} + \cancel{x} - \cancel{x} - \cancel{x^2} + \cancel{x^2} + x^3 - \cancel{1}}{1+x}$$

$$= \frac{x^3}{1+x} \checkmark$$

$$b) p^2 - x + q = 0 \quad -1, 3.$$

$$\therefore \text{sum, } -1+3 = -\frac{b}{a}$$

$$2 = \frac{1}{p}$$

$$\therefore p = \frac{1}{2}$$

$$\text{product} \quad -3 = \frac{c}{a}$$

$$-3 = \frac{q}{p}$$

$$-3p = q$$

$$-3 \cdot \frac{1}{2} = q$$

$$\therefore q = \underline{\underline{-\frac{3}{2}}}$$

$$c) 4^x - 5(2^{x+1}) + 16 = 0$$

$$(2^x)^2 - 5 \cdot 2^x \cdot 2^1 + 16 = 0$$

$$\text{let } u = 2^x \rightarrow u^2 - 10u + 16 = 0$$

$$(u-8)(u-2) = 0$$

$$\therefore u = 8 \text{ or } 2.$$

$$\therefore 2^x = 8 \text{ or } 2$$

$$\therefore x = 3 \text{ or } 1$$

$$d) i) f(x) = \frac{x}{x-1}$$

$$f\left(\frac{1}{2}\right) - f(-2)$$

$$\frac{\frac{1}{2}}{\frac{1}{2}-1} - \frac{-2}{-2-1}$$

$$\frac{\frac{1}{2}}{-\frac{1}{2}} - \frac{-2}{-3}$$

$$-1 - \frac{2}{3}$$

$$= \underline{\underline{-1\frac{2}{3}}}$$

Ques 6 cont...

$$ii) f(x) = \frac{x}{x-1}$$

$$\begin{array}{l} \text{even, } f(x) = f(-x) \\ \text{odd, } f(x) = -f(-x) \end{array}$$
$$\frac{x}{x-1} = \frac{-x}{-x-1}$$

~~$$\frac{x}{x-1} = \frac{-x}{-x-1}$$~~

$$\frac{x}{x-1} = \frac{-x}{-(x+1)}$$
$$= \frac{x}{x+1}$$

$$\text{LHS} = f(x) \quad \text{RHS} = f(-x)$$

$$\text{now } f(x) \neq f(-x)$$

$$f(x) \neq -f(-x)$$

$\therefore$ neither odd nor even

$$e) (1 - \cos \theta)(1 + \sec \theta) = \sin \theta \cdot \tan \theta$$

$$1 + \sec \theta - \cos \theta - \cos \theta \cdot \sec \theta$$

$$= 1 + \frac{1}{\cos \theta} - \cos \theta - \cancel{\cos \theta} \cdot \frac{1}{\cancel{\cos \theta}}$$

$$= \frac{1}{\cos \theta} - \cos \theta$$

$$\frac{1 - \cos^2 \theta}{\cos \theta}$$

$$= \frac{+\sin^2 \theta}{\cos \theta} = \frac{+\sin \theta \cdot \sin \theta}{\cos \theta}$$

$$= +\tan \theta \cdot \sin \theta$$

$\therefore \text{RHS}$

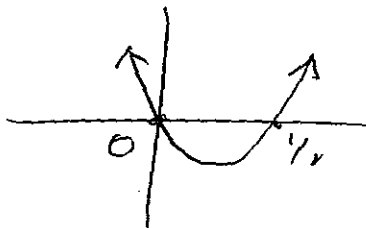
Ques 7

a) $2x^2 \geq x$

$$2x^2 - x \geq 0$$

$$x(2x-1) \geq 0$$

$$x = 0, \frac{1}{2}$$



$$\therefore x \leq 0 \cup x \geq \frac{1}{2}$$

b) $\int 6 - x^{-3} dx$

$$= 6x - \frac{6x^{-2}}{-2} + C$$

$$= 6x + \frac{6}{2x^2} + C$$

$$= 6x + \frac{3}{x^2} + C$$

OR $6x + 3x^{-2} + C$

c) i) angle sum = $180(n-2)$

$$= 180(11-2)$$

$$= 180 \times 9$$

$$= 1620^\circ$$

ii) $\therefore L = \frac{1620^\circ}{11}$

$$= 147\frac{3}{11}$$

$$x 60 = \frac{180}{11}$$

$$= 147^\circ 16'$$

d) $5x^2 + 2x - 3$

$$= A(x+1)^2 + B(x+1) + C$$

$$RHS = A(x^2 + 2x + 1) + Bx + B + C$$

$$= Ax^2 + 2Ax + A + Bx + B + C$$

$$\therefore A = 5$$

$$2A + B = 2$$

$$\therefore 10 + B = 2$$

$$B = -8$$

and $A + B + C = -3$

$$\therefore 5 - 8 + C = -3$$

$$-3 + C = -3$$

$$\therefore C = 0$$

$$\therefore A = 5, B = -8, C = 0$$

e) $x^2 + 4x - 8y + 12 = 0$

$$x^2 + 4x = 8y - 12$$

$$x^2 + 4x + 4 = 8y - 12 + 4$$

$$(x+2)^2 = 8y - 8$$

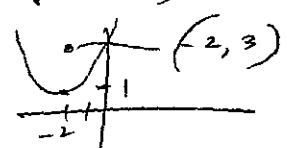
$$(x+2)^2 = 8(y-1)$$

$$\therefore \text{Vertex} = (-2, 1)$$

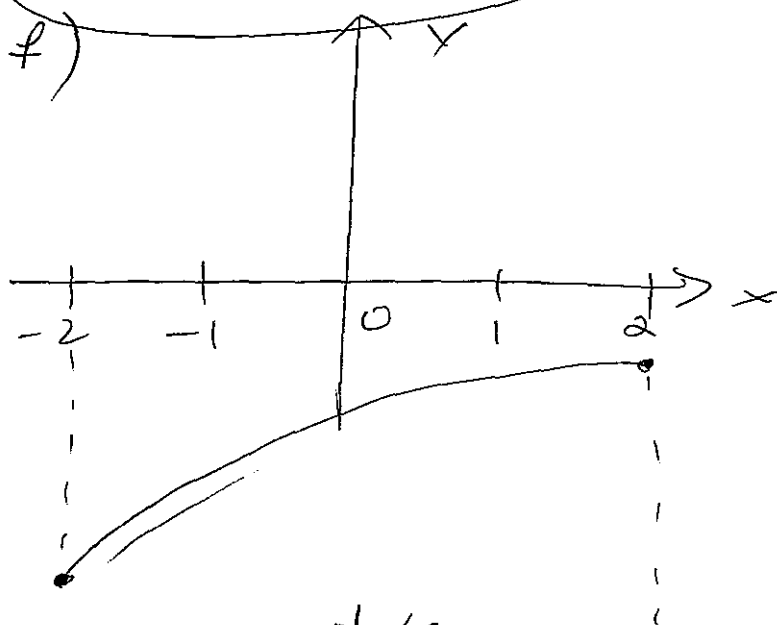
$$4a = 8$$

$$a = 2$$

$$\therefore \text{Focus} = (-2, 3)$$

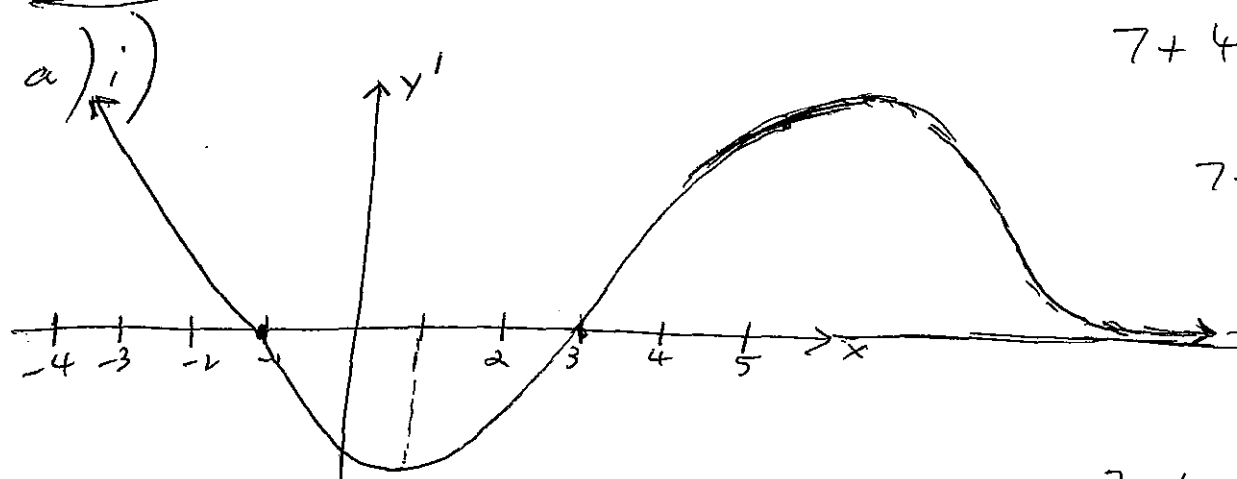


Ques 7 cont...



y is negative
gradient is positive
concave down.

Ques 8)



$$7 + 4 \cdot \frac{8}{27} - 3 \times \frac{16}{81}$$

$$7 + \frac{32}{27} - \frac{48}{81}$$

$$7 + \frac{96 - 48}{81}$$

$$7 - 4 - 3$$

$$7 + 32 - 48$$

ii) $f'(x) < 0$; $-1 < x < 3$

iii) $f'(x) \rightarrow 0$ as $x \rightarrow \infty$

iii) pt of inflex

$$12x(2-3x) = 0$$

$$x = 0, x = \frac{2}{3}$$

$$f''\left(\frac{1}{2}\right) = \oplus$$

$$f''(1) = \ominus$$

b) $y = 7 + 4x^3 - 3x^4$

i) $y' = 12x^2 - 12x^3$
 $= 12x^2(1-x)$

at pnts, $y' = 0$

$$12x^2(1-x) = 0$$

$$\therefore x = 0, 0, 1$$

ii) $y'' = 24x - 36x^2$
 $= 12x(2-3x)$

$f''(0), y'' = 0 \therefore$ poss. pt. of inflex

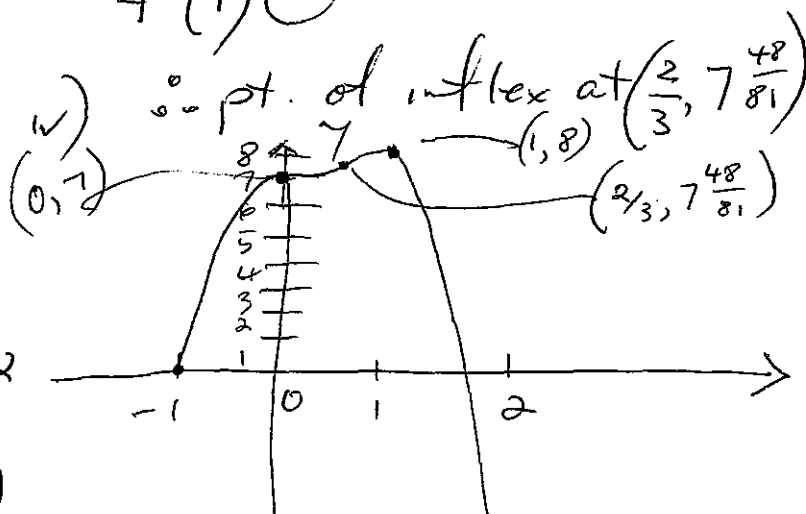
$f''\left(-\frac{1}{2}\right), f''\left(\frac{1}{2}\right)$

$$f''\left(-\frac{1}{2}\right) = \ominus \oplus \rightarrow \ominus$$

$$f''\left(\frac{1}{2}\right) = \oplus \oplus \rightarrow \oplus$$

$\therefore$ pt of inflexion at $(0, 7)$

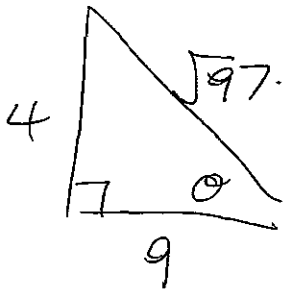
$x = 1, y'' < 0 \therefore$ max pt at $(1, 8)$



v) $y = 8$ ($x = 1$)

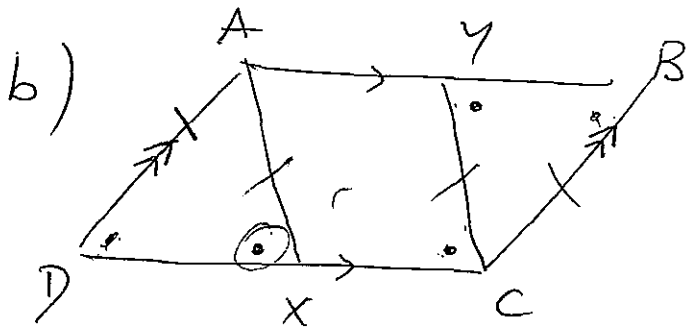
Ques 9

a) $\tan \theta = \frac{-4}{9}$ $\cos \theta = \frac{9}{\sqrt{97}}$
4th quad.



$$\begin{array}{r} 5 \overline{) 47} \\ \underline{45} \\ 20 \\ \underline{20} \\ 0 \end{array}$$

$\therefore \sin \theta = \frac{-4}{\sqrt{97}}$



i) opposite angles of a parallelogram are equal $\therefore \angle AXD = \angle CYB$

ii) $\angle CYB = \angle BYC$ (equal L's of isos. $\triangle$)
 $\angle CXA = \angle BYC$ (alt. L's $AB \parallel CD$)
 $\angle AXD =$

$BC = AD$ (opp. sides of $\parallel$ gram)
now $BC = CY = AX$ (given)
so $AX = AD$.

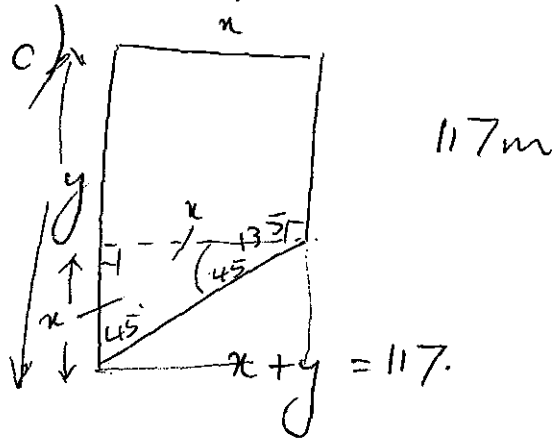
iii) In $\triangle$'s ADC , $\triangle CBY$.
 $AD = BC$ (opp. sides of $\parallel$ gram)
 $AX = CY$ (given)
 $\angle AXD = \angle CYB$ (180 - 2 equal angles)

$\therefore \triangle ADC \equiv \triangle CBY$
(2 sides and included angle test)

iv) $AY \parallel XC$ ($ABCD$ is $\parallel$ gram)
 $YC \parallel AX$ ($\angle CYB = \angle AXD$)
corres. L's

$\therefore AYCX$ is 4 sided figure with 2 pairs of parallel sides
 $\therefore$ parallelogram.

Ques 9 cont.



$$\therefore y = 117 - x$$

$$\begin{aligned} A &= x \times y - \frac{x \times x}{2} \\ &= x(117 - x) - \frac{x^2}{2} \\ &= 117x - x^2 - \frac{x^2}{2} \end{aligned}$$

$$A = 117x - \frac{3x^2}{2}$$

$$\text{ii) } A' = 117 - \frac{6x}{2}$$

$$\text{At pt. } A' = 0$$

$$117 - 3x = 0$$

$$3x = 117$$

$$x = 39$$

$$\therefore y = 117 - 39 = 78$$

$$\therefore y = 78 = 2 \times 39 \text{ i.e. } y = 2x$$

$$A'' = -3$$

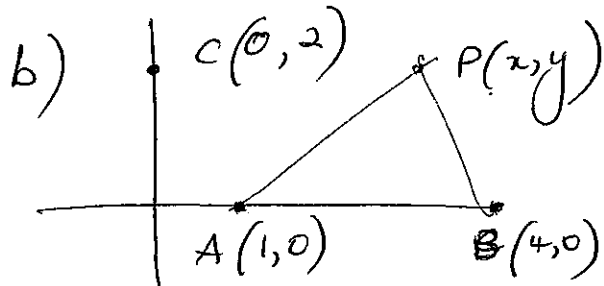
$\therefore$ max area

$$\text{when } x = 39, y = 78 \text{ or } y = 2x$$

Ques 10

$$a) \frac{1-x^2}{x-1} = \frac{(1-x)(1+x)}{x-1}$$

$$= -(1+x)$$



Now $CB = 2CA$.

$$CB = \sqrt{2^2 + 4^2}$$

$$= \sqrt{4 + 16}$$

$$= \sqrt{20} = 2\sqrt{5}$$

$$CA = \sqrt{2^2 + 1^2}$$

$$= \sqrt{5}$$

~~$\therefore CA = 2$~~

$$CB = 2CA$$

$$(2\sqrt{5} = 2 \times \sqrt{5})$$

ii) $PB = 2PA$.

$$PB = \sqrt{(x-4)^2 + (y-0)^2}$$

$$PA = \sqrt{(x-1)^2 + (y-0)^2}$$

$$(x-4)^2 + y^2 = 4((x-1)^2 + y^2)$$

$$x^2 - 8x + 16 + y^2 = 4(x^2 - 2x + 1 + y^2)$$

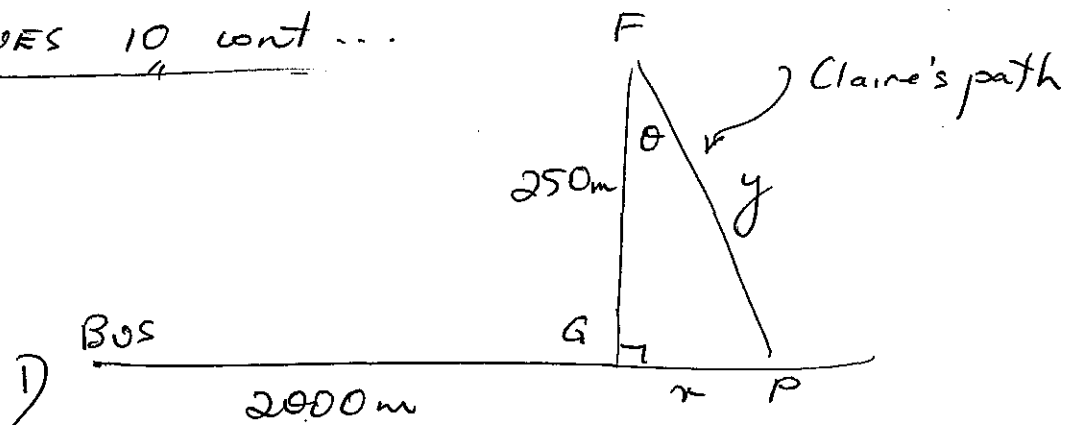
$$x^2 - 8x + 16 + y^2 = 4x^2 - 8x + 4 + 4y^2$$

$$3x^2 - 3y^2 = 12$$

circle centre (0,0)

QUES 10 cont ...

c)



$$i.) S = \frac{D}{T}$$

$$T = \frac{D}{S}$$

$$\text{let } GP = x$$

$$\text{now } \tan \theta = \frac{x}{250}$$

$$x = 250 \tan \theta$$

$$\text{also } y^2 = x^2 + 250^2$$

$$y = \sqrt{x^2 + 62500}$$

$$= \sqrt{62500 \tan^2 \theta + 62500}$$

$$15 \text{ m/s} = 54000 \text{ m/h}$$

$$4 \text{ m/s} = 14400 \text{ m/h}$$

$$T_{\text{bus}} = \frac{2000 + 250 \tan \theta}{54000} \quad h = \frac{4 + \tan \theta}{216} \text{ hours}$$

$$T_{\text{claire}} = \frac{\sqrt{62500 \tan^2 \theta + 62500}}{14400} = \frac{250 \sqrt{\tan^2 \theta + 1}}{14400}$$

$$= \frac{25 \sqrt{\sec^2 \theta}}{1440}$$

$$= \frac{25 \sec \theta}{1440} \text{ hours.}$$

$$\therefore T_{\text{bus}} = \frac{4 + \tan \theta}{216} \quad ; \quad T_{\text{claire}} = \frac{5 \sec \theta}{288}$$